

Project Report: Stock Analyzer

Course: Fundamentals of AIML (BYOP)

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1. Problem Statement & Significance

The Problem: The financial markets are flooded with "noise"—volatile price movements that make it difficult for retail investors to identify actual trends. Emotional trading often leads to losses.

Why it Matters: There is a need for an objective, **Rule-Based Expert System** that can process historical data and provide logical buy/sell signals based on proven financial heuristics, removing human emotion from the decision-making process.

2. AIML Approach

My approach was to build a symbolic AI system that mimics a financial analyst's reasoning.

- **Knowledge Representation:** I used a structured Knowledge Base (CSV) containing time-series data for three major Indian stocks: **TCS, INFY, and SBI**.
 - **Feature Engineering:** Raw "Close" prices were preprocessed into **Short-term (3-day)** and **Long-term (5-day)** Moving Averages to extract trend features.
 - **Inference Engine:** I implemented a logic unit that monitors "State Changes." A prediction is only triggered when a "Crossover" occurs, representing a shift in market momentum.
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3. Key Decisions & Architecture

- **Multi-Ticker Support:** I decided to move from a single-stock script to a multi-entity system. This required adding a **Metadata Layer (Ticker column)** to the CSV and a **Filtering Layer** to the Python code.

- **Modular Coding:** I broke the code into distinct functions (load_data, apply_heuristics, make_prediction) to ensure the system is scalable and easy to debug.
 - **Heuristic Selection:** I chose the "Golden Cross" and "Death Cross" heuristics because they are mathematically sound and widely respected in algorithmic trading.
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4. Challenges Faced

- **Environment Pathing Issues:** A major hurdle was the FileNotFoundError. The Python interpreter was looking in the terminal's root directory instead of the script's folder.
 - *Solution:* I implemented **Dynamic Absolute Pathing** using the os module to ensure the Knowledge Base is always found.
 - **Logic Edge Cases:** Initially, the system predicted "Golden Cross" every time the Short MA was above the Long MA.
 - *Solution:* I added **State-Based Memory** logic (comparing latest vs previous rows) so the AI only flags the *exact moment* a trend reverses.
 - **Data Scarcity:** Calculating a 5-day average requires at least 5 rows of data.
 - *Solution:* I added a **Validation Gate** that checks the length of the filtered dataset before running the inference engine.
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5. Testing & Results

I validated the system using three controlled datasets to ensure 100% accuracy in classification:

- **Bullish Test:** Verified that TCS triggered a "Golden Cross" when price momentum spiked.
 - **Bearish Test:** Verified that INFY triggered a "Death Cross" during a simulated market drop.
 - **Neutral Test:** Verified that SBI remained "Neutral" during a steady, non-crossing growth phase.
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6. What I Learned

Through this BYOP, I gained hands-on experience in the **AI Development Life Cycle**. I learned that data cleaning and path management are just as important as the algorithm itself. Most importantly, I understood how **Rule-Based AI** can provide clear, explainable logic for complex fields like FinTech, where transparency is vital.